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It is a closely related problem to approximate $\|M\|_{p \rightarrow q}$ in polynomial time. In the regime $2 < q < \infty$, $p > 1$, the known polynomial time algorithms give upper and lower bounds that differ by a power of N (cf. [GMU] and references therein). For some values of p, q in this range, it is known to be NP -hard to approximate $\|M\|_{p \rightarrow q}$ within a constant factor – see [BV], [BBHKSZ], and [BGGLT]. The methods in these papers may lead to similar results for the large value problem.

Because of these computational difficulties, there is no meaningful numerical evidence supporting the main conjectures about large values of Dirichlet polynomials. This is different from the situation with the Riemann hypothesis itself. There is a lot of numerical evidence supporting the Riemann hypothesis – for instance, we know that the first 10^9 zeroes of the Riemann zeta function have real part equal to $1/2$ (EDIT: Reference). In contrast, we are not able to check the large value conjectures numerically even for $N = 200$.

There is not currently a conjectural picture of how well polynomial time algorithms can approximate $LV_{M,\ell^p}(\lambda)$. If M is allowed to vary among all $T \times T$ matrices, then it looks reasonable to conjecture that it is NP -hard to approximate the function $LV_{M,\ell^p}(N^\sigma)$ to within a factor T^γ , for some $\gamma > 0$. But there is no clear conjecture about the sharp value of the exponent γ .

Proving bounds for $LV_{M,\ell^p}(\lambda)$ is not just hard in the worst case, it is also hard in the average case. If M is a random $T \times N$ matrix, then Proposition 3.2 shows that with high probability, M obeys an essentially optimal large value estimate. But if we are given a particular matrix M that was sampled from the random distribution in Proposition 3.2, it is still hard to prove that M obeys strong large value estimates. In computer science, this is called the problem of certifying large value estimates for M . It is closely related to the problem of distinguishing an honest random matrix M from a matrix that has a planted structure which causes $LV_{M,\ell^p}(\lambda)$ to be much larger.

I think the problem of certifying large value estimates for random matrices is closely parallel to the large value problem for Dirichlet polynomials. The two problems have been studied independently, but the methods and bounds that are known for the two problems are closely parallel, as we will see in the next few sections.

The problem of certifying large value estimates for random matrices is part of the field of average case computational complexity. For a broad class of related problems in the field, the best current method is called the sum of squares hierarchy. The limits of this method are well understood in a number of cases. It is not fully understood what large value estimates for a random matrix M can be certified by the sum of squares method, but there was a lot of recent progress in [DHPT]. Based on recent developments in the field, there is good evidence that the sum of squares method cannot certify that a random matrix obeys the large value estimates in Proposition 3.2 or Conjecture 3.3 or 3.4.

Experts in the field consider it to be plausible that the sum of squares method is the best polynomial time algorithm for a broad class of problems in average case computational complexity, including certifying large value estimates for random matrices. So there is a plausible conjectural picture about what large value estimates can be certified in polynomial time. In this picture, almost every random matrix obeys very strong large value estimates, but these strong large value estimates cannot be certified by a polynomial time algorithm.

After this brief discussion of computational complexity, let us return to the question why the large value problem for Dirichlet polynomials is hard. Recall that M_{Dir} is the $T \times N$ matrix that encodes Dirichlet polynomials. We would like to prove bounds for $LV_{M_{Dir},\ell^\infty}(\lambda)$ for all N and all $T = N^{O(1)}$. However, let us set a more modest goal and try to prove bounds when $N = 10^3$ and